

Teaching Exam

General Science

Units & Measurement, Motion

Lecture No. - 1



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Topics to be Covered



Topic

Units and Measurement

Topic

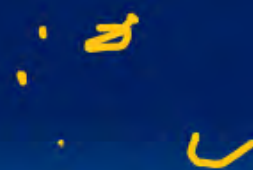
Topic

Topic

Topic

राशि(QUANTITY)

- जिसको संख्या के रूप में प्रकट किया जा सके उसे राशि कहा जाता है।/ A QUANTITY THAT CAN BE EXPRESSED IN THE FORM OF A NUMBER IS CALLED A QUANTITY.



Physical Quantities-

10 Kg → mass



TEACHING
WALLAH

- ❖ All those quantities which can be measured./वे सभी मात्राएं जिन्हें मापा जा सकता है।
- ❖ All quantities in terms of which laws of physics can be expressed and which can be measured are called Physical Quantities.
- ❖ The quantity which can be measured by an instrument and which describes the laws of physical world./वह मात्रा जिसे एक यंत्र द्वारा मापा जा सकता है और जो भौतिक संसार के नियमों का वर्णन करती है।
- ❖ Example: length, mass, time, temperature, force, pressure, work./उदाहरण: लंबाई, द्रव्यमान, समय, तापमान, बल, दबाव, काम।

भौतिक राशियाँ

PHYSICAL QUANTITY

1. SCALAR QUANTITY

अदिश राशि

2. VECTOR QUANTITY

सदिश राशि

✓ Scalar Quantities/अदिश राशि-

- ❖ Are not dependent on direction/दिशा पर निर्भर नहीं हैं
- ❖ Can be described only by magnitude/केवल परिमाण द्वारा व्यक्त
किया जा सकता है
- ❖ If only magnitude is required to describe a physical quantity,
then it is a scalar quantity.
- ❖ Some of the examples are Charge, distance, area, speed,
time, temperature, density, volume, work, power, energy,
pressure, potential etc

Vector Quantities/सदिश राशि -

- ❖ It is dependent on both direction as well as magnitude./यह दिशा और परिमाण दोनों पर निर्भर करता है।
- ❖ If magnitude as well as direction is required to describe any physical quantity, then it is a vector quantity.
- ❖ Some of the examples are
- ❖ displacement, velocity, acceleration, force, momentum, torque, gravitational intensity etc.

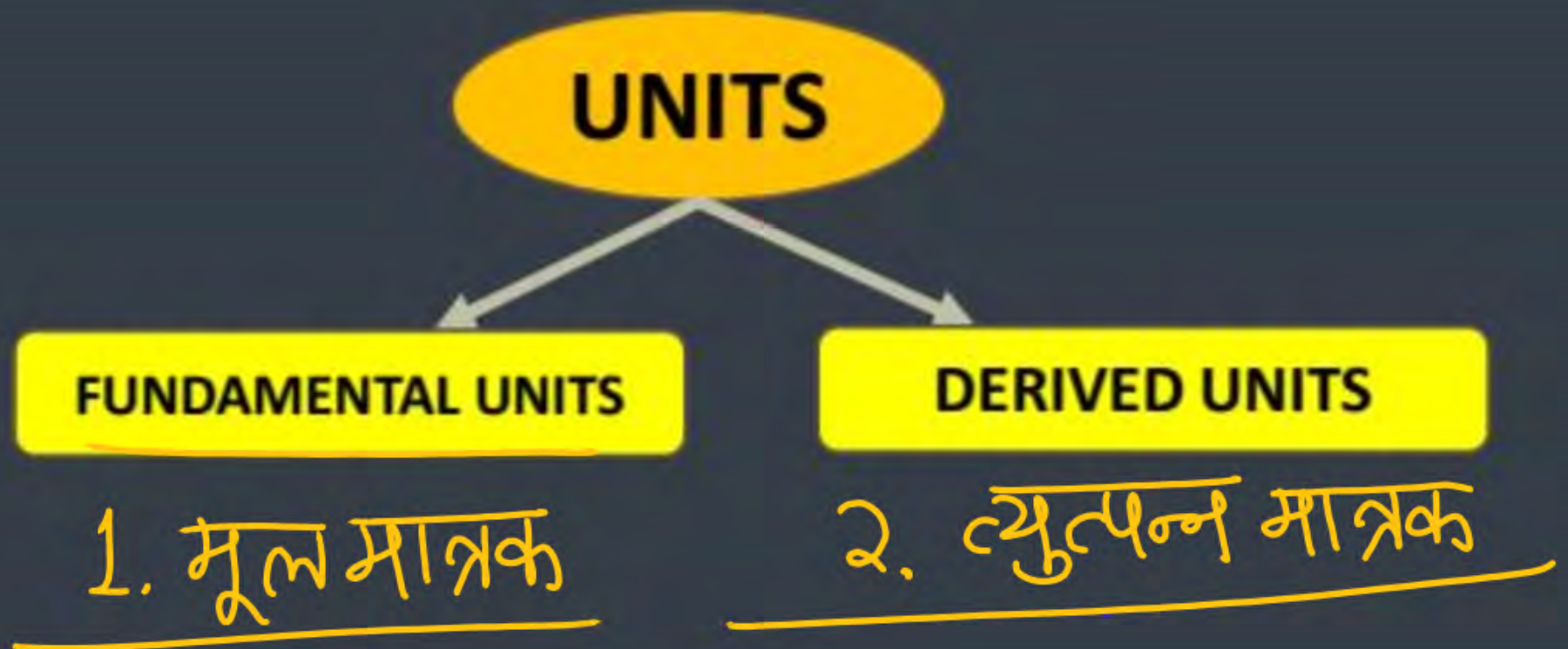
SCALAR અદિશ	VECTOR સદિશ
<u>DISTANCE</u> દૂરી	DISPLACEMENT વિસ્થાપન
SPEED ચાલ	VELOCITY વેગ
TIME સમય ક્ષેત્રફલ, આયતન	ACCELERATION ત્વરણ
AREA, VOLUME, TEMPERATURE તાપમાન	FORCE બળ
PRESSURE, દાબ કાર્ય, ઊર્જા, શક્તિ	MOMENTUM સંવેગ
WORK, ENERGY & POWER ધારા, વિભવ	WEIGHT ભાર
CURRENT, ELECTRIC POTENTIAL, RESISTANCE પ્રતિરોધ	TORQUE બલ આધૂર્ણ

Difference between Scalar and Vector

Scalar	Vector
It has <u>magnitude only</u> .	It has <u>magnitude as well as direction</u> .
It changes <u>with the change in magnitude</u> .	It changes <u>with the change in magnitude or change in direction</u> .
It can be added or subtracted using <u>basic rule of algebra</u> .	It can be added or subtracted using <u>geometrical method (such as, triangle law of vectors addition or parallelogram law of vectors addition)</u> .

UNITS -

TOOLS USING
WHICH CAN
MEASURE
PHYSICAL
QUANTITIES
EFFECTIVELY.



मूल मात्रक (Fundamental Units) –

- ❖ किसी भौतिक राशि को व्यक्त करने के लिए प्रयुक्त वे मात्रक जो अन्य मात्रकों से स्वतंत्र होते हैं। / The units used to express a physical quantity which are independent of other units.
- ❖ इनकी संख्या **7** है।
- ❖ जैसे – मीटर, सेकेण्ड, किलोग्राम, एम्पियर, मोल, केल्विन, कैंडेला। / E.g. Meters, Seconds, Kilograms, Ampere, Mole, Kelvin, Candela.

मूल मात्रक (Fundamental Units) –

- ❖ The quantity which is independent of other physical quantities. In mechanics, mass, length and time are called fundamental quantities. Units of these fundamental physical quantities are called Fundamental units.
- ❖ वह राशि जो अन्य भौतिक राशियों से स्वतंत्र होती है, उसे मौलिक राशि कहा जाता है। यांत्रिकी में, द्रव्यमान, लंबाई और समय को मौलिक राशियाँ कहा जाता है। इन मौलिक भौतिक राशियों की इकाइयाँ मौलिक इकाइयाँ कहलाती हैं।

व्युत्पन्न मात्रक (Derived Units) -

❖ दो या दो से अधिक मूल मात्रकों के माध्यम से व्यक्त किया जाता है | Expressed through two or more basic units.

❖ जैसे -

m/sec
Speed

m/sec^2
acceleration
त्वरण

$kg\ m/sec^2$
Force

Quantity (भौतिक राशि)	Derived Unit (व्युत्पन्न इकाई)	SI Symbol	Formula (सूत्र)
<u>Area (क्षेत्रफल)</u>	Square meter (वर्ग मीटर)	m^2 ✓	<u>Length × Breadth</u> ($m \times m$)
Volume (आयतन)	<u>Cubic meter (घन मीटर)</u>	m^3	Length × Breadth × Height ($m \times m \times m$)
Speed/Velocity (गति/वेग)	<u>Meter per second (मीटर प्रति सेकंड)</u>	m/s	Distance ÷ Time (m/s)
Acceleration (त्वरण)	<u>Meter per second squared (मीटर/सेकंड²)</u>	m/s^2	Velocity ÷ Time ($m/s/s$)
Force (बल)	<u>Newton (न्यूटन)</u>	N	Mass × Acceleration ($kg \times m/s^2$)
Work/Energy (कार्य/ऊर्जा)	<u>Joule (जूल)</u> ↓	J	Force × Distance ($N \times m$)

UNIT SYSTEMS-

1. CGS or French system

cm - gram - second

2. MKS or Metric system

meter - kg - second

3. FPS or British system

Foot - Pound - second

4. SI or international system of unit

- (i) C.G.S system: In this system, the unit of length is centimetre, the unit of mass is gram and the unit of time is second.
- (ii) F.P.S system: In this system, the unit of length is foot, the unit of mass is pound and the unit of time is second.
- (iii) M.K.S: In this system, the unit of length is metre, unit of mass is kg and the unit of time is second.
- (iv) S.I System: This system is an improved and extended version of M.K.S system of units. It is called international system of unit.

FUNDAMENTAL UNIT-

Which do not depend on any other physical quantity./जो किसी अन्य भौतिक राशि पर निर्भर नहीं करते हैं।

FUNDAMENTAL UNIT-

Fundamental Quantities		SI Unit
1. Length	→	1. Meter (m)
2. Mass	→	2. Kilogram (kg)
3. Time		3. Second (S) ✓
4. Electric current		4. Ampere (A) ✓
5. Temperature		5. Kelvin (K) ✓
6. Amount of substance		6. Mol (mol) ✓
7. Luminous Intensity		7. Candela (cd) ✓

Table of Fundamental Units

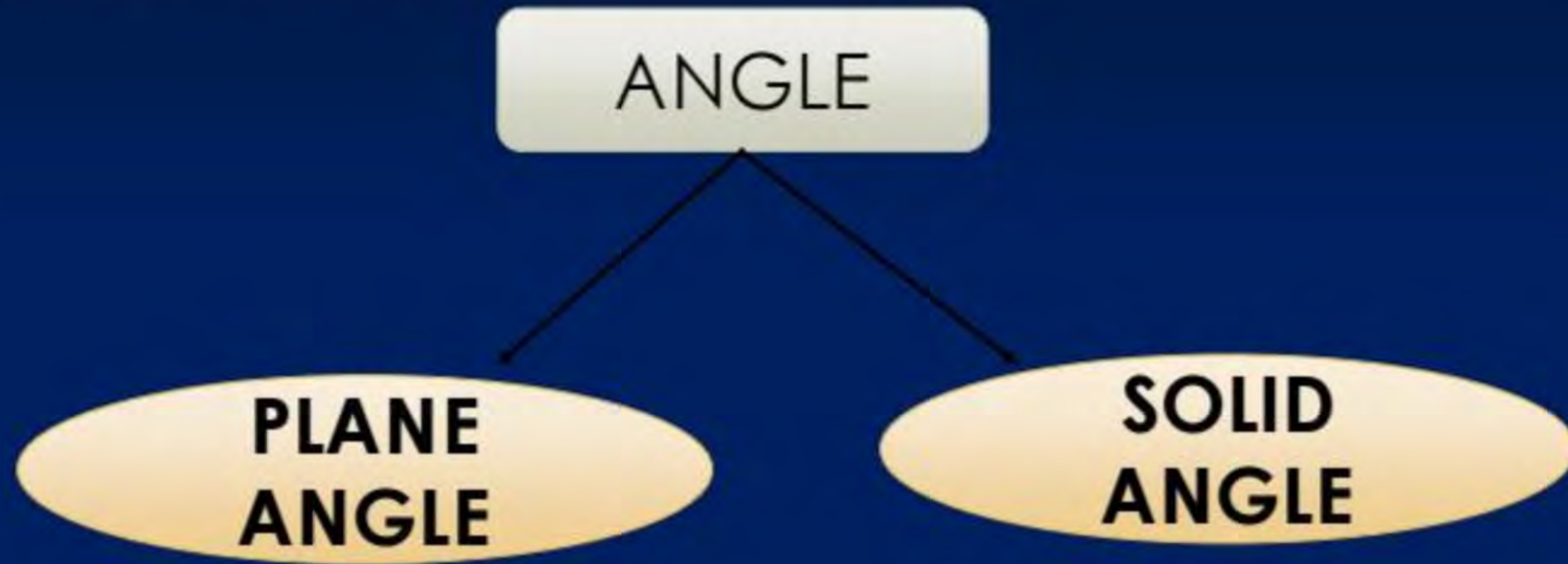
Sr. No.	Name of Physical Quantity	Unit	Symbol
1	Length	Metre	m ✓
2	Mass	Kilogram	Kg ✓
3	Time	Second	s ✓
4	Temperature	Kelvin	K ✓
5	Electric Current	Ampere	A ✓
6	Luminous Intensity	Candela	Cd ✓
7	Quantity of Matter	Mole	<u>mol</u>

सम्पूरक मात्रक

Table of Supplementary unit

Sr. No	Name of Physical Quantity	Unit	Symbol
1	<u>Plane angle</u> समतल कोण	<u>Radian</u> →	<u>rad</u>
2	<u>Solid angle</u> ठोस कोण	<u>Steradian</u> →	<u>sr</u>

SUPPLEMENTARY UNIT-



-
रेडियन/radians

- स्टेरेडियन/steredian

Symbol	Prefix	Multiplier	Symbol	Prefix	Multiplier
D	<u>Deci</u>	<u>10^{-1}</u>	da	<u>deca</u>	<u>10^1</u>
c	<u>centi</u>	<u>10^{-2}</u>	h	<u>hecto</u>	<u>10^2</u>
m	<u>milli</u>	<u>10^{-3}</u>	k	<u>kilo</u>	<u>10^3</u>
μ	<u>micro</u>	<u>10^{-6}</u>	M	<u>mega</u>	<u>10^6</u>
n	<u>nano</u>	<u>10^{-9}</u>	G	<u>giga</u>	<u>10^9</u>

P	Pico	10^{-12}	T	tera	10^{12}
f	femto	10^{-15}	P	Pecta	10^{15}
a	atto	10^{-18}	E	exa	10^{18}

याद करिये

Some Important Units of Length:

- (i) 1 micron = 10^{-6} m = 10^{-4} cm
- (ii) 1 angstrom = $1 \text{ \AA}$ = 10^{-10} m = 10^{-8} cm ✓
- (iii) 1 fermi = 1 fm = 10^{-15} m
- (iv) 1 Light year = 1 ly = 9.46×10^{15} m ✓
- (v) 1 Parsec = 1 pc = 3.26 light year ✓

Some conversion factor of mass:

$$1 \text{ Kilogram} = 2.2046 \text{ pound}$$

$$1 \text{ Pound} = 453.6 \text{ gram}$$

$$1 \text{ kilogram} = 1000 \text{ gram}$$

$$1 \text{ milligram} = 1/1000 \text{ gram} = 10^{-3} \text{ gram}$$

$$1 \text{ centigram} = 1/100 \text{ gram} = 10^{-2} \text{ gram}$$

$$1 \text{ decigram} = 1/10 \text{ gram}$$

$$1 \text{ quintal} = 100 \text{ kg}$$

$$1 \text{ metric ton} = 1000 \text{ kilogram}$$

Sr. No.	Physical Quantity	Formula	Dimensions	Name of S.I unit
1	Force ✓	Mass $\times$ acceleration	$[M^1L^1T^{-2}]$	Newton (N)
2	Work ✓	Force $\times$ distance	$[M^1L^2T^{-2}]$	Joule (J)
3	Power ✓	Work / time	$[M^1L^2T^{-3}]$	Watt (W)
4	Energy (<u>all form</u>) ✓	Stored work	$[M^1L^2T^{-2}]$	Joule (J)
5	<u>Pressure, Stress</u>	Force/area	$[M^1L^{-1}T^{-2}]$	Nm^{-2}
6	<u>Momentum</u>	Mass $\times$ velocity	$[M^1L^1T^{-1}]$	$Kgms^{-1}$
7	<u>Moment of force</u>	Force $\times$ distance	$[M^1L^2T^{-2}]$	Nm
8	<u>Impulse</u>	Force $\times$ time	$[M^1L^1T^{-1}]$	Ns
9	<u>Strain</u>	Change in dimension / Original dimension	$[M^0L^0T^0]$	No unit

10	<u>Modulus of elasticity</u>	Stress / Strain	$[M^1L^{-1}T^{-2}]$	Nm^{-2}
11	Surface energy ✓	Energy / Area	$[M^1L^0T^{-2}]$	Joule/m ²
12	Surface Tension ✓	Force / Length	$[M^1L^0T^{-2}]$	N/m ✓
13	Co-efficient of viscosity ✓	Force × Distance / Area × Velocity	$[M^1L^{-1}T^{-1}]$	N/m ²
14	Moment of inertia ✓	Mass × (radius of gyration) ²	$[M^1L^2T^0]$	Kg-m ²
15	Angular Velocity ✓	Angle / time	$[M^0L^0T^{-1}]$	Rad.per sec
16	Frequency ✓	1/Time period	$[M^0L^0T^{-1}]$	Hertz
17	Area	Length × Breadth	$[M^0L^2T^0]$	Metre ² ✓
18	Volume	Length × breadth × height	$[M^0L^3T^0]$	Metre ³ ✓

19	Density	Mass/ volume	$[M^1L^{-3}T^0]$	Kg/m^3 ✓
20	Speed or velocity	Distance/ time	$[M^0L^1T^{-1}]$	m/s ✓
21	Acceleration	Velocity/time	$[M^0L^1T^{-2}]$	m/s^2 ✓
22	Pressure	Force/area	$[M^1L^{-1}T^{-2}]$	N/m^2 ✓

Derived unit -

Which are made up of fundamental physical quantity. जो मौलिक भौतिक मात्रा से बने होते हैं।

$$\text{Area} = l \times b = \text{m}^2$$

- ❖ तरंगदैर्घ्य - एंगस्ट्रम
- ❖ परम ताप - केल्विन
- ❖ समुद्र की गहराई - फैदम
- ❖ गुप्त ऊष्मा - जूल प्रति किग्रा.
- ❖ चुंबकीय क्षेत्र - गौस
- ❖ ज्योति फ्लक्स - ल्यूमेन
- ❖ पराध्वनिक गति - मैक

- ❖ Wavelength - Angstrom
- ❖ Absolute Temperature - Kelvin
- ❖ sea depth - fathom
- ❖ Latent heat - joule per kg.
- ❖ magnetic field - gauss
- ❖ Light Flux - Lumen
- ❖ supersonic speed - Mach

- ❖ लेंस की क्षमता - डाइ ऑप्टर
- ❖ संवेग/आवेग - न्यूटन सेकण्ड
- ❖ पृष्ठ तनाव - न्यूटन प्रति मीटर
- ❖ विभवांतर - वोल्टे
- ❖ जड़त्व आघर्षण - किग्रा. वर्ग मी.
- ❖ खगोलीय दूरी - प्रकाश वर्ष
- ❖ श्यानता - न्यूटन सेकण्ड मी. ⁻²
- ❖ Lens Power - Diopter
- ❖ momentum/impulse - newton second
- ❖ surface tension - newton per meter
- ❖ potential difference - volt
- ❖ Moment of Inertia - kg. Sq.m.
- ❖ astronomical distance - light year
- ❖ Viscosity - Newton second meter. ⁻²

- ❖ चंबकीय फ्लक्स - वेबर, मैक्सवेल
- ❖ विद्युत् क्षेत्र तीव्रता - न्यूटन प्रति कूलम्ब
- ❖ ज्योति तीव्रता - कैण्डेला
- ❖ गुरुत्वीय त्वरण - मीटर प्रति सेकण्ड²
- ❖ वायुमंडलीय दाब - बार
- ❖ चंबकीय तीव्रता - टेसला
- ❖ चंबकीय प्रेरण - गाउस
- ❖ ठोस कोण - स्टेरेडियन
- ❖ कोणीय वेग - रेडियन प्रति सेकण्ड

- ❖ Magnetic Flux - Weber, Maxwell
- ❖ Electric Field Intensity - Newton per Coulomb
- ❖ Luminous Intensity - Candela
- ❖ Gravitational acceleration - meter per second²
- ❖ atmospheric pressure - bar
- ❖ Magnetic Intensity - Tesla
- ❖ Magnetic Induction - Gauss
- ❖ solid angle - steradian
- ❖ angular velocity - radian per second

Table 1.2 Some units retained for general use (Though outside SI)

Name	Symbol	Value in SI Unit
minute	min	60 s
hour	h	60 min = 3600 s
day	d	24 h = 86400 s
year	y	365.25 d = 3.156×10^7 s
degree	°	$1^\circ = (\pi/180)$ rad
litre	L	$1 \text{ dm}^3 = 10^{-3} \text{ m}^3$
tonne	t	10^3 kg
carat	c	200 mg
bar	bar	$0.1 \text{ MPa} = 10^5 \text{ Pa}$
curie	Ci	$3.7 \times 10^{10} \text{ s}^{-1}$
roentgen	R	$2.58 \times 10^{-4} \text{ C/kg}$
quintal	q	100 kg
barn	b	$100 \text{ fm}^2 = 10^{-28} \text{ m}^2$
are	a	$1 \text{ dam}^2 = 10^2 \text{ m}^2$
hectare	ha	$1 \text{ hm}^2 = 10^4 \text{ m}^2$
standard atmospheric pressure	atm	$101325 \text{ Pa} = 1.013 \times 10^5 \text{ Pa}$

नाम	प्रतीक	SI मात्रक के पदों में मान
मिनट	min	60 s
घंटा	h	60 min = 3600 s
दिन	d	24 h = 86400 s
वर्ष	y	365.25 d = 3.156×10^7 s
डिग्री	°	$1^\circ = (\pi/180)$ rad
लिटर	L	$1 \text{ dm}^3 = 10^{-3} \text{ m}^3$
टन	t	10^3 kg
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क्यूरी	Ci	$3.7 \times 10^{10} \text{ s}^{-1}$
रॉजन	R	$2.58 \times 10^{-4} \text{ C kg}^{-1}$
क्विंटल	q	100 kg
बार्न	b	$100 \text{ fm}^2 = 10^{-28} \text{ m}^2$
आर	a	$1 \text{ dam}^2 = 10^2 \text{ m}^2$
हेक्टर	ha	$1 \text{ hm}^2 = 10^4 \text{ m}^2$
मानक वायुमंडलीय दाब	atm	$101\,325 \text{ Pa} = 1.013 \times 10^5 \text{ Pa}$

Name	Major contribution/discovery	Country of Origin
<u>Archimedes</u>	Principle of buoyancy; Principle of the lever ✓	Greece
Galileo Galilei	Law of inertia ✓	Italy
<u>Christiaan Huygens</u>	Wave theory of light ✓	Holland
Isaac Newton ✓	Universal law of gravitation; Laws of motion; Reflecting telescope	U.K.
<u>Michael Faraday</u> ✓	Laws of electromagnetic induction ✓	U.K.
<u>James Clerk Maxwell</u>	Electromagnetic theory; <u>Light-an electromagnetic wave</u>	U.K.
Heinrich Rudolf Hertz ✓	Generation of <u>electromagnetic waves</u>	Germany
J.C. Bose	Ultra short radio waves ✓	India
W.K. Roentgen ✓	X-rays ✓	Germany
J.J. Thomson ✓	Electron ✓	U.K.
Marie Sklodowska Curie ✓	Discovery of <u>radium and polonium</u> ; <u>Studies on natural radioactivity</u>	Poland
Albert Einstein ✓	Explanation of photoelectric effect; <u>Theory of relativity</u>	Germany

Name	Major contribution/discovery	Country of Origin
Victor Francis Hess ✓✓	<u>Cosmic radiation</u>	Austria
R.A. Millikan ✓	<u>Measurement of electronic charge</u>	U.S.A.
<u>Ernest Rutherford</u>	<u>Nuclear model of atom</u>	New Zealand
<u>Niels Bohr</u>	<u>Quantum model of hydrogen atom</u>	Denmark
<u>C.V. Raman</u>	<u>Inelastic scattering of light by molecules</u>	India
<u>Louis Victor de Borglie</u>	<u>Wave nature of matter</u>	France
<u>M.N. Saha</u>	<u>Thermal ionisation</u>	India
S.N. Bose ✓	<u>Quantum statistics</u> ✓	India
Wolfgang Pauli ✓	<u>Exclusion principle</u> ✓	Austria
Enrico Fermi ✓	<u>Controlled nuclear fission</u> ✓	Italy
Werner Heisenberg ✓✓	<u>Quantum mechanics; Uncertainty principle</u> ✓	Germany

Put
It
General
Science

~~Miss~~

Paul Dirac ✓	Relativistic theory of electron; <u>Quantum statistics</u>	U.K.
Edwin Hubble ✓	<u>Expanding universe</u>	U.S.A.
<u>Ernest Orlando Lawrence</u>	<u>Cyclotron</u>	U.S.A.
James Chadwick ✓	<u>Neutron</u>	U.K.
Hideki Yukawa ✓	<u>Theory of nuclear forces</u>	Japan
Homi Jehangir Bhabha ✓	<u>Cascade process of cosmic radiation</u>	India
Lev Davidovich Landau ✓	<u>Theory of condensed matter; Liquid helium</u>	Russia
S. Chandrasekhar ✓	<u>Chandrasekhar limit, structure and evolution of stars</u>	India
John Bardeen ✓	<u>Transistors; Theory of super conductivity</u> ✓	U.S.A.
C.H. Townes ✓	<u>Maser; Laser</u> ✓	U.S.A.
Abdus Salam ✓	<u>Unification of weak and electromagnetic interactions</u>	Pakistan



Technology	Scientific principle(s)
Steam engine →	<u>Laws of thermodynamics</u>
Nuclear reactor →	<u>Controlled nuclear fission</u>
Radio and Television →	<u>Generation, propagation and detection of electromagnetic waves</u>
Computers →	<u>Digital logic</u>
Lasers →	<u>Light amplification by stimulated emission of radiation</u>
Production of ultra high magnetic fields ≫	<u>Superconductivity</u>
Rocket propulsion ✓	<u>Newton's laws of motion</u>
Electric generator ✓	<u>Faraday's laws of electromagnetic induction</u>
Hydroelectric power ✓	<u>Conversion of gravitational potential energy into electrical energy</u>

Aeroplane	Bernoulli's principle in fluid dynamics
Particle accelerators ✓	Motion of charged particles in electromagnetic fields
Sonar ✓	Reflection of ultrasonic waves
Optical fibres ✓	Total internal reflection of light
Non-reflecting coatings ✓	Thin film optical interference
Electron microscope ✓	Wave nature of electrons
Photocell ✓	Photoelectric effect
Fusion test reactor (Tokamak) ✓	Magnetic confinement of plasma
Giant Metrewave Radio Telescope (GMRT) ✓	Detection of cosmic radio waves
Bose-Einstein condensate	Trapping and cooling of atoms by laser beams and magnetic fields.

Let put

Table 1.3 Fundamental forces of nature

Name	Relative strength	Range	Operates among
Gravitational force	10^{-39}	Infinite	All objects in the universe
Weak nuclear force	10^{-13}	Very short, Sub-nuclear size ($\sim 10^{-16}\text{m}$)	Some elementary particles, particularly electron and neutrino
Electromagnetic force	10^{-2}	Infinite	Charged particles
Strong nuclear force	1	Short, nuclear size ($\sim 10^{-15}\text{m}$)	Nucleons, heavier elementary particles

Base quantity		
	Name	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Rest विराम

यदि किसी वस्तु की स्थिति
समय के साथ न बदले।

If an object does not change
its position with r. to time.

Motion गति

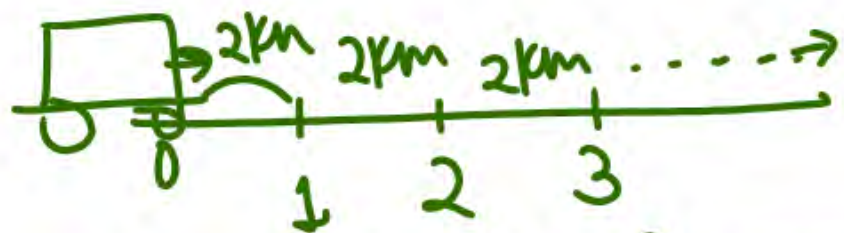
यदि वस्तु की स्थिति, समय के
साथ बदले।

If an object change its
position w.r.t. time.

Motion

Uniform motion

एक समान गति

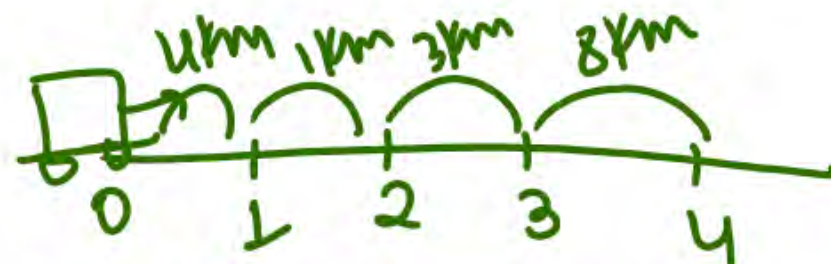


यदि वस्तु समान समय में समान दूरी तय करें।

If an object cover equal distance in equal interval of time.

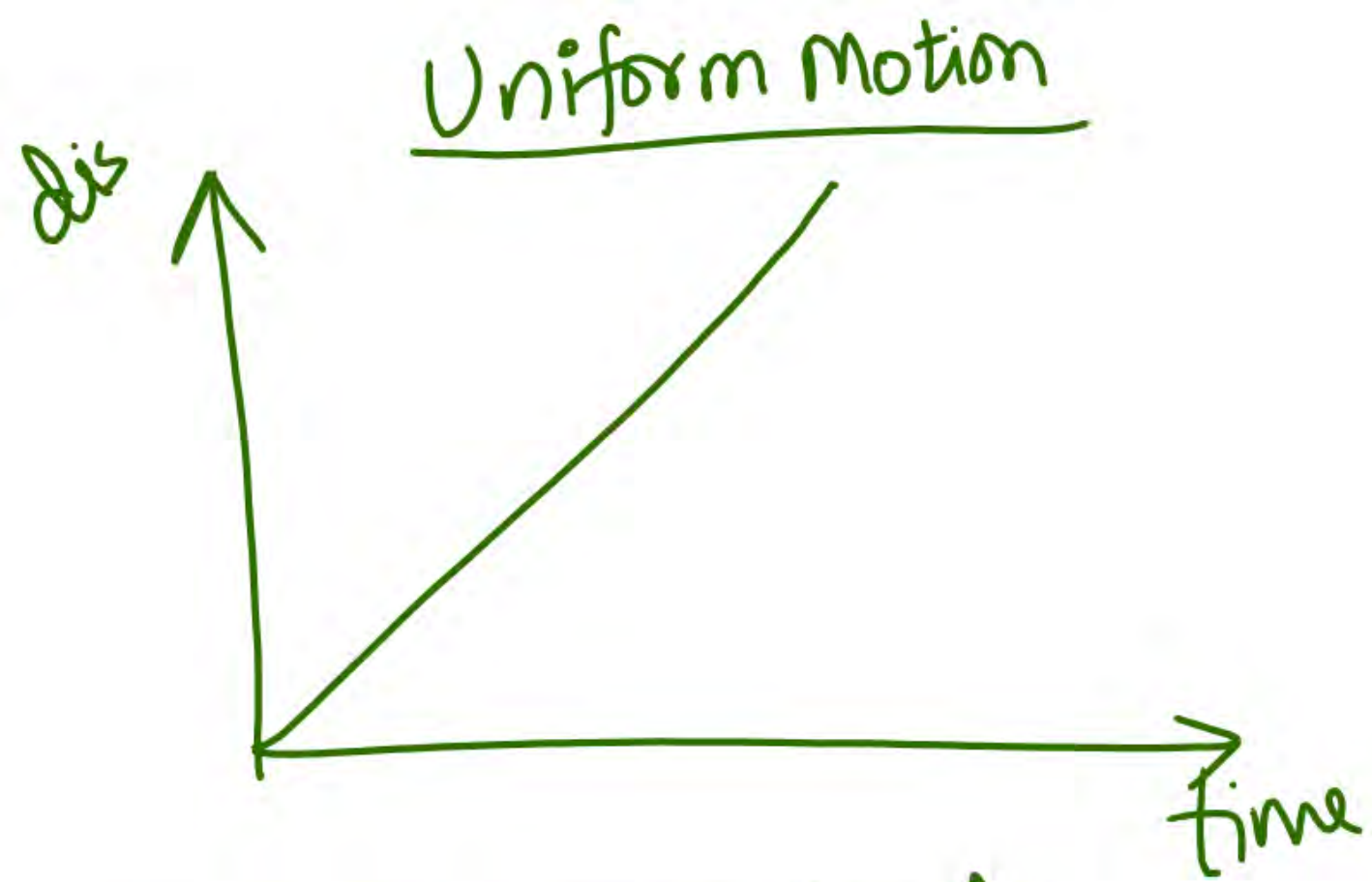
Non-uniform motion

असमान गति



यदि वस्तु समान समय में असमान दूरी तय करें।

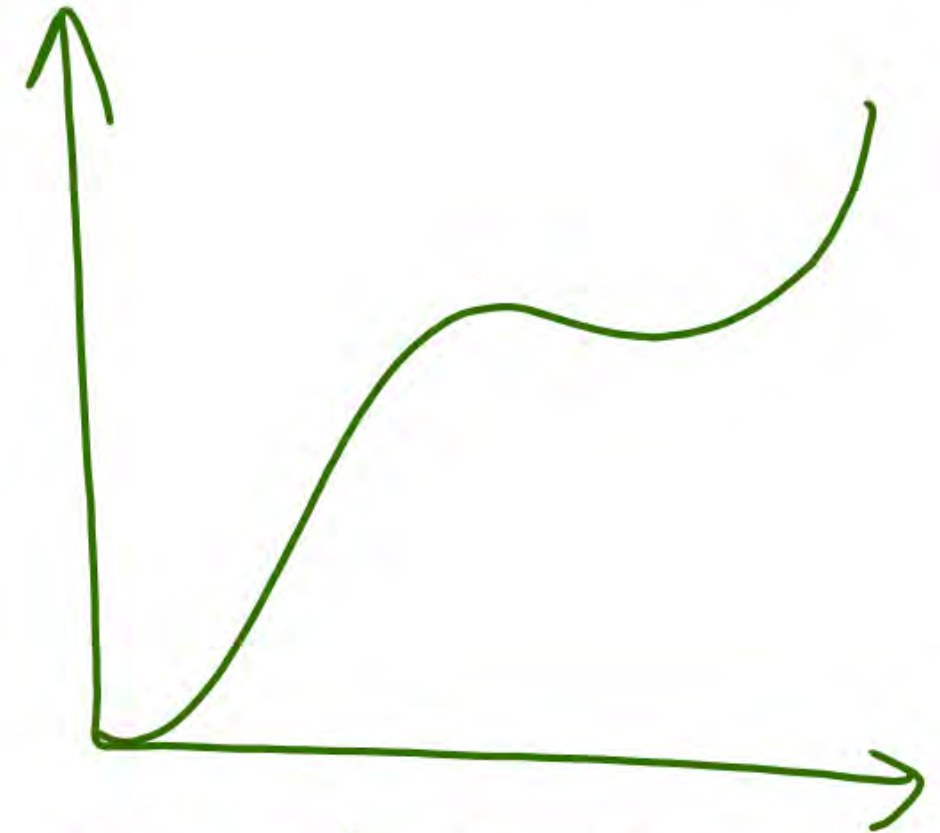
If an object cover unequal distance in equal interval of time.



$\Rightarrow$ Velocity constant

$\Rightarrow$ $a = 0$

Nonuniform Motion  **TEACHING WALLAH**



$\Rightarrow$ velocity change

$\Rightarrow$ $a \neq 0$ ✓

Types of Motion

- 1) Rectilinear Motion - स्करेखीय गति → एक सीधी रेखा में गति।
- 2) Circulatory Motion - वृत्तीय गति → वृत्ताकार मार्ग में गति।
- 3) Periodic/Oscillatory Motion - आवधिक/दोलनी गति - To & fro motion
जैसे - पेंडुलम में गति





2 mins Summary



TEACHING
WALLAH

Topic

One

HW

Revision

Learn

Topic

Two

Topic

Three

Topic

Four

Topic

Five

Motion
Units

THANK YOU